

TURRET OS: Multi-Layered Graph and Rule-AST Architecture Rescuing Insider Threat Detection Against Out-of-Distribution Adversaries

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ABSTRACT

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1. Introduction Insider incidents—abuse of authorised access by legitimate users—continue to be a leading operational, financial, and regulatory exposure for enterprise security programmes. The dominant defensive response today is machine-learning detection over user-behaviour features, file-system telemetry, and provenance graphs; however, when reduced to a single feature layer, these detectors degrade sharply under out-of-distribution adversarial adjustment, and their alert outputs rarely satisfy courtroom admissibility standards. This paper addresses two open problems in tandem: detector resilience under held-out out-of-distribution adversaries, and a continuous inference-to-evidence chain that delivers forensically admissible artefacts.

Prior research clusters in four strands. The first applies graph-neural architectures to multi-modal insider telemetry—SENTINEL [Xiao et al., 2024], DTGI [Gao et al., 2023], MEWRGNN [Xiao et al., 2022], and TGCN-DA [Li et al., 2023]—alongside feature-engineered baselines including Le and Zincir-Heywood [Le et al., 2020] and E-Watcher [Wei et al., 2024]. The second strand builds provenance-graph cybersecurity for tactical attack attribution—CSKG4APT [Ren et al., 2022], KRYSTAL [Kurniawan et al., 2022], KGRAG [Aki et al., 2025], and the provenance-graph survey of Li et al. [Li et al., 2021]—but typically terminates outputs at alert generation. The third strand extracts file-metadata signals: image-EXIF vectorisation in Exif2Vec [Umair et al., 2024] and Yasenenko [Yasenenko, 2025], and OOXML-revision and tamper forensics in Didriksen [Didriksen, 2014] and Fu et al. [Fu et al., 2015]. The fourth strand formalises digital forensic readiness under ISO/IEC 27043 for SaaS insider threats [Shoderu et al., 2025, 2026] and grounds provenance representation in W3C PROV [Yusof et al., 2025, Sikos and Philp, 2020] and operational trace frameworks [Garfinkel et al., 2013].

Several gaps remain. (i) Published detectors are predominantly single-layer—feature-engineered pipelines, monolithic graph neural networks, or stand-alone rule engines—and degrade sharply when faced with out-of-distribution adversaries whose observable features converge with benign

baseline traffic [Wei et al., 2024, Xiao et al., 2024, Gao et al., 2023]. (ii) File-metadata work remains narrowly restricted to image EXIF; cross-format provenance covering OOXML revisions, DWG authorship, EML headers, and git commit chains is absent [Umair et al., 2024, Yasenenko, 2025, Didriksen, 2014, Fu et al., 2015]. (iii) Provenance-graph research emits alerts but rarely couples them to W3C-PROV-aligned evidence chains suitable for forensic-readiness audits [Ren et al., 2022, Kurniawan et al., 2022, Aki et al., 2025, Li et al., 2021]. (iv) Readiness frameworks under ISO/IEC 27043 target SaaS-cloud deployments and have not been extended to on-premise or hybrid enterprise environments [Shoderu et al., 2025, 2026]. (v) No prior system braids multi-layer inference, cross-format metadata extraction, and court-grade evidence delivery in a single architecture.

Against these gaps we propose TURRET OS,¹ a four-layer architecture that fuses a cross-format metadata-harvest layer (L1), a Neo4j knowledge-graph layer (L2), a rule-AST inference layer (L3), and a multi-layer GraphSAGE inference layer (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chain-integrity-protected, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes—Identification, Collection, Acquisition, Preservation, Analysis, Presentation, Chain of Custody, and Integrity Verification. Under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only a 4.12% drop, while single-layer feature detectors collapse to AUC 0.7363 with a 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (35 users, 90 days) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.5% FPR of 0.8936, and MCC 0.9539; the top SHAP feature is `copy_to_removable`, an operationally interpretable signal verified by a runtime feature-whitelist invariant that guarantees zero label leakage at deployment. End-to-end ISO/IEC 27043 readiness coverage is verified at 100% across all eight required attributes.

The remainder of the paper is organised as follows. Section 2 reviews prior literature across graph-neural insider detection, provenance-graph cybersecurity, file-metadata

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¹Project repository: <https://github.com/ali-Hamza817/Turret-OS>

forensics, and ISO/IEC 27043 readiness. Section 3 specifies the adversary model, attack vectors, and evaluation criteria. Section 4 presents the TURRET OS architecture layer by layer. Section 5 reports the implementation stack on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 platform. Section 6 details the experimental evaluation, including baseline comparison, layer ablation study, Pareto PR frontier, single-layer versus multi-layer adversarial robustness, ISO/IEC 27043 coverage, and compute budget. Section 7 discusses architectural and operational implications. Section 8 outlines limitations and future work. Section 9 concludes.

2. Literature Review

Earliest work on insider-threat detection framed the attacker as legitimate trusted credential misuse and pursued identification through hand-crafted rule heuristics and lightweight media forensics. [Garfinkel et al. \[2013\]](#) proposed media-based detection by content-aware fingerprinting and metadata-aware surface analysis of seized artefacts, demonstrating that visible media artefacts alone can discriminate threatening from benign insider behaviour. [Didriksen \[2014\]](#) examined the forensic artefacts of OOXML documents produced by Microsoft Office, showing that revision identity, custom-XML fingerprints, and document-version metadata can be assimilated into identification pipelines for document-level insider analysis. [Fu et al. \[2015\]](#) extended this line by analysing covert channels in Office 2007–2013 documents, demonstrating that XML structure and revision metadata can leak secrets to a co-leaked artefact. Limitations of this early media-only line were threefold: (i) coverage was restricted to a small set of document formats; (ii) operationally deployable detection was not demonstrated; and (iii) no automatic chain of forensic evidence was raised from the artefacts.

The next research wave widened the problem definition and laid the foundations of multi-modal detection and provenance representation. [Mavroeidis et al. \[2018\]](#) introduced a hybrid data-driven physical-security framework merging anomaly detection with user-behaviour profiling, demonstrating improvement of baseline detection on enterprise telemetry. [Le and Zincir-Heywood \[2019\]](#) and [Le et al. \[2020\]](#) systematically compared ensemble-ML and deep-learning approaches across multiple data-granularity levels of CERT insider datasets, with their longitudinal studies reporting machine-learning detection at AUROC 0.92 and F1 0.88 with time-to-detect of 14 minutes. [Homoliak et al. \[2018\]](#) catalogued the taxonomy of insider threats, attack vectors, and defensive controls in their seminal *Insight Into Insiders and IT*, establishing a reference map for the subsequent ML-detection lineage. [Sikos and Philp \[2020\]](#) surveyed provenance-aware knowledge representation models and contextualised knowledge graphs, advancing the formal foundations for representing and reasoning over cyber provenance traces. [Li et al. \[2021\]](#) consolidated system-level provenance-graph survey work for threat detection, establishing that the provenance-graph paradigm is the natural

substrate for post-hoc forensic reasoning over multi-modal telemetry. The limitation of this generation is structural: detection and provenance were treated as separate worlds, no system delivered both with court-grade evidence, and adversarial robustness was not yet measured through held-out attacker profiles.

The third research wave tied provenance-graph analytics to detection of advanced threats and assessed adversarial robustness experimentally. [Ren et al. \[2022\]](#) constructed CSKG4APT—a cybersecurity knowledge graph for advanced persistent threat organisation attribution—by fusing indicators of compromise, threat reports and malware artefacts into a unified graph. [Kurniawan et al. \[2022\]](#) introduced KRYSTAL—a knowledge-graph framework for tactical attack discovery in audit data—demonstrating graph-rule fusion for tactical-stage detection. [Xiao et al. \[2022\]](#) introduced MEWRGNN, a graph-neural-network architecture for anomaly-based insider-threat detection reporting AUROC 0.94 and F1 0.87. [Gao et al. \[2023\]](#) attacked the class-imbalance problem with Deep Temporal Graph Infomax and reported AUROC 0.91 and F1 0.86. [Li et al. \[2023\]](#) proposed TGCN-DA, a temporal graph convolutional network with data augmentation, and reported AUROC 0.95 and F1 0.89. [Xiao et al. \[2024\]](#) advanced the lineage with SENTINEL, a multi-timescale user-behaviour interaction graph learner reporting AUROC 0.93 and F1 0.88. [Wei et al. \[2024\]](#) reported E-Watcher with 98.48% accuracy and 99.23% F1 on CERT r4.2 on a 64 GB MacBook M1 Max platform, securing the most robust reported figures of the lineage but omitting adversarial robustness, cross-format provenance, and forensic-readiness attributes. [Mitra et al. \[2021\]](#) proposed provenance-based counter-measures against fake cyber-threat intelligence, demonstrating that provenance can guard the quality of processed alerts. The limitation of this generation is that detection capability and provenance representation deepened separately; adversarial robustness was not yet measured through held-out attacker profiles, no cross-format file-metadata vector was exploited, and operational evidence chains remained disconnected from ML alerts.

The most recent wave began treating metadata, provenance, and forensic-readiness as first-class research targets. [Umair et al. \[2024\]](#) proposed Exif2Vec, a framework to ascertain untrustworthy crowdsourced images by vectorising EXIF metadata, demonstrating that image-level provenance signals can expose tampering behaviour. [Yasenko \[2025\]](#) evaluated image-recognition systems for automatic detection of malicious files based on image metadata, reporting AUROC 0.76 on a curated image set. [Aki et al. \[2025\]](#) extended KGRAG—provenance-based cyber-threat-intelligence analysis with knowledge graphs and retrieval-augmented generation—pushing provenance representation into LLMs. [Yusof et al. \[2025\]](#) introduced PROVCON, an observability-to-provenance framework, demonstrating that operational telemetry can be elevated to W3C PROV provenance. [Shoderu et al. \[2025\]](#) and [Shoderu et al. \[2026\]](#) formalised digital forensic readiness under ISO/IEC 27043 for

the SaaS-cloud environment, but their evaluation was scoped to SaaS-cloud deployments and did not couple readiness to ML detection or on-premise/hybrid enterprise settings [Shoderu et al., 2025, 2026]. The remaining open gap of this generation is the absence of a unifying architecture that fuses cross-format metadata extraction, multi-layer joint inference, ISO/IEC 27043 readiness, and held-out attacker profiles within a single deployable system.

Synthesising the limitations of the four research waves—narrow format coverage of media forensics, separation of detection and provenance, absence of held-out adversarial regimes, and missing on-premise forensic-readiness coupling—we propose TURRET OS,² a four-layer architecture that fuses cross-format metadata harvest (L1), a Neo4j knowledge graph (L2), a rule-AST inference engine (L3), and multi-layer GraphSAGE inference (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chained, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes. End-to-end detection with the multi-layer stack attains ROC-AUC 0.9997, F1 @ 0.5% FPR of 0.8774, and MCC 0.9770 on the D3 TSEC corpus; under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only 4.12% drop, while single-layer L1 feature detectors collapse to AUC 0.7363 with 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (D5, 35 users over 90 days, proactively tagged by Security Operations analysts) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.8936, and MCC 0.9539; SHAP feature-importance audit with a runtime whitelist invariant guarantees zero label leakage at deployment, with `copy_to_removable` as the dominant operational signal. ISO/IEC 27043 readiness coverage is verified at 100% across all eight attributes; computational budget is profiled with wall-clock timestamps on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 deployment platform. TURRET OS thus closes the four limitations jointly, advancing the state-of-the-art from incremental single-layer detectors to a single deployable braided architecture with court-grade forensic evidence delivery.

3. Threat Model and Problem Formulation

3.1. Adversary Profiles

TURRET OS evaluates adversary profiles across two corpora. On the established CERT-style insider-threat lineage corpus (D3 TSEC) the embedded adversary baseline consists of five synthetic insider scenarios generated over 500 users, 365 days, and 141,657 user-day activity records, with a 1.68% positive rate [Wei et al., 2024, Xiao et al., 2024, Gao et al., 2023]. On the 35-user, 90-day D5 Synthetic-Tenant Pilot (2,431 activity records) a held-out OOD_LAB_EXFILTRATOR profile is injected into the generator without appearing in the training split; this deliberately out-of-distribution scenario functions as a true hold-out evaluation. Across both corpora, four canonical attacker models

anchor the adversarial evaluation track in Table V: Mimicry, Metadata-Strip, Identity-Proxy (GHOST_AUTHOR), and OOD Credential-Phishing. The OOD Credential-Phishing adversary operates during business hours using stolen credentials with low feature novelty, identity-proxy reuse, and removable-media transfers—the regime in which classical novelty-driven insider detectors collapse.

3.2. Attack Vectors and Evasion Strategies

Four evasion strategies are exercised. Mimicry approximates benign-feature distributions to defeat single-layer feature detectors. Metadata-Strip Evasion removes authorship and revision-id signals from OOXML [Didriksen, 2014, Fu et al., 2015], PDF, DWG, EML, git, and image artefacts [Umair et al., 2024, Yassenko, 2025]; this vector specifically targets L1 metadata-only detectors. Identity-Proxy Injection substitutes a clean user's credential surface for the attacker's actions, collapsing naive identity-conditioned rules. OOD Credential-Phishing combines stolen legitimate credentials, business-hours operation, no metadata-stripping, low feature novelty, identity-proxy reuse, and removable-media transfers—the regime in which novelty-, off-hours-, and metadata-driven insider detectors systematically collapse [Ren et al., 2022, Kurniawan et al., 2022]. The four vectors span the limiting cases any deployable insider-threat detector must survive end-to-end.

3.3. Forensic Readiness Requirements

Detection must be coupled to a forensic-evidence chain admissible under ISO/IEC 27043 [Shoderu et al., 2025, 2026]. The architectural target is 100% attribute coverage across the eight mandated attributes: Identification (unique record and alert UUIDs), Collection (documented harvest-pipeline provenance), Acquisition (SHA-256 + BLAKE3 hash verification), Preservation, Analysis (SHAP + GN-Explainer provenance subgraphs), Presentation (analyst-UI workbench export format), Chain of Custody (timestamps and operator-signature tracking), and Integrity Verification (Ed25519 signature over the Merkle root) [Shoderu et al., 2025]. This requirement elevates the threat model from "raise an alert" to "raise an alert admissible as evidence under a forensic readiness audit".

3.4. Detection and Robustness Evaluation Criteria

Detection is reported on three primary metrics: ROC-AUC, F1 @ 0.5% FPR (the precision-critical operating point for low-false-positive deployability), and Matthews Correlation Coefficient (MCC). Calibration is performed via `CalibratedClassifierCV` (isotonic, cv=3); operating thresholds are swept via Youden's J statistic. Target on D3 TSEC: **ROC-AUC 0.9997, F1@0.5%FPR 0.8774, MCC 0.9770**, with the operating threshold pinned at the 0.5% FPR constraint to match the deployment. Target on the D5 OOD_LAB_EXFILTRATOR hold-out: **ROC-AUC 0.9928, F1 0.8936, MCC 0.9539**. Adversarial robustness is bound by **AUC-drop ≤ 5%** against each of the four attacker vectors in §3.2. Cross-layer attribution is bound by a runtime safety invariant: the top-5 SHAP feature importances on the chronological train/test split must

²Project repository: <https://github.com/ali-Hamza817/Turret-OS>

Table 1

Literature Review Summary Table Mapping Key Findings and Structural Limitations Across Prior Work.

Study	Year	Key Findings	Limitations
Didriksen [2014]	2014	Forensic analysis of OOXML artefacts establishing that revision-identity fields, custom-XML fingerprints, and document-version metadata can be assimilated into identification pipelines.	Single document format only; no coupled detection model; no held-out adversary evaluation.
Fu et al. [2015]	2015	Forensic analysis of Microsoft Office 2007–2013 documents demonstrating that XML structure and revision metadata can be exploited; declared useful for covert-channel detection in document artefacts.	Covert-channel extraction orientation; no continuous insider-detection deployment; no held-out attacker evaluation.
Mavroeidis et al. [2018]	2018	Data-driven physical-security framework unifying anomaly detection with user-behaviour profiling for enterprise-telemetry insider analysis.	Detection and provenance treated as separate worlds; no ISO/IEC 27043 readiness alignment; no held-out attacker profile.
Li et al. [2021]	2021	Survey of threat detection and investigation with system-level provenance graphs consolidating the provenance-graph paradigm.	Detection and provenance surveyed as separate strands; no cross-format metadata integration; no held-out adversary regime.
Kurniawan et al. [2022]	2022	KRYSTAL knowledge-graph framework for tactical-attack discovery in audit data using graph-rule fusion.	Alert generation decoupled from forensic-chain emission; no held-out attacker profiles; no cross-format metadata layer.
Ren et al. [2022]	2022	CSKG4APT cybersecurity knowledge graph for advanced-persistent-threat organisation attribution over indicators-of-compromise, threat reports, and malware artefacts.	Attribution-focused; alerts not coupled to W3C PROV evidence; on-premise coverage absent.
Xiao et al. [2022]	2022	MEWRGNN architecture for anomaly-based insider threat detection over user-action graphs.	Single-layer architecture; adversarial robustness not measured; no forensic-readiness integration.
Gao et al. [2023]	2023	DTGI using Deep Temporal Graph Infomax to handle class-imbalance in insider-threat detection.	Delivery decoupled from provenance chain; single-layer GNN without held-out attackers.
Li et al. [2023]	2023	TGCN-DA temporal graph convolutional network with data augmentation for insider-threat detection.	Augmentation over-fit risk on population singletons; no cross-format metadata; no held-out attacker profile.
Wei et al. [2024]	2024	E-Watcher insider-threat monitoring and detection framework; reporting ROC-AUC 0.9848 on its own benchmark and headline metric accuracy 98.48%.	Single-layer; adversarial robustness not reported; no cross-format metadata fusion; no forensic-readiness audit.
Umair et al. [2024]	2024	Exif2Vec framework vectorising EXIF metadata to ascertain untrustworthy crowdsourced images.	Restricted to image EXIF format; no extension to OOXML/DWG/EML/git metadata; no detection–provenance coupling.
Xiao et al. [2024]	2024	SENTINEL multi-timescale user-behaviour interaction graph learner for insider detection.	Single-layer GNN; missing cross-format metadata layer; no ISO/IEC 27043 readiness alignment.
Aki et al. [2025]	2025	KGRAG provenance-based cyber-threat-intelligence analysis aligning provenance graphs with retrieval-augmented generation.	CTI-oriented; no continuous internal-detection SLI; no cross-format metadata layer; no forensic-readiness audit.
Yasenko [2025]	2025	Evaluation framework for image-recognition systems for malicious-file detection based on image metadata.	Image-format scope only; no extension to document metadata; no detection–provenance alignment.
Yusof et al. [2025]	2025	PROVCON observability-to-provenance framework constructing cyberattack provenance at W3C PROV fidelity.	Detection layer absent; no held-out attackers; no cross-format metadata integration.
Shoderu et al. [2025]	2025	Digital forensic readiness framework under ISO/IEC 27043 for insider-mitigation in SaaS-cloud environments.	SaaS-cloud deployment only; no on-premise/hybrid scope; no detection–provenance coupling.
Shoderu et al. [2026]	2026	Formalised ISO/IEC 27043 forensic-readiness framework for insider mitigation across SaaS deployments.	SaaS-cloud scope only; ML detection not coupled.

belong to a fixed operational whitelist (access_novelty_score, metadata_stripped, off_hours_multiplier, identity_proxy, copy_to_removable, n_file_accesses, outbound_email, access_hour) guaranteeing zero label leakage at deployment.

4. TURRET OS Architecture

The pipeline has four logically distinct layers plus a multi-layer joint-inference stage. Each layer is independently testable and contributes a clearly bounded signal into both the anomaly score and the forensic-evidence bundle.

4.1. Layer 1 — Cross-Format Metadata Harvest

Ingests OOXML (legacy and current), DOCX (legacy and current), XLSX, PPTX, PDF, DWG, EML, EPIC payload, bitmap images, and git commits [[Didriksen, 2014](#), [Fu et al., 2015](#)]. Extracted fields include author, creator, lastModifiedBy, revision_id, rsid_originator, rsid_revisions, parent_rsid_chain, total_edit_count, last_modification_date, git_user_email, and creation_timestamp. The L1 store exposes twelve derived feature columns: access_novelty_score, metadata_stripped, off_hours_multiplier, identity_proxy, copy_to_removable, n_file_accesses, outbound_email, access_hour, revshare, comm_share_by_hour, write_open_share, and bytes_used_relative

[Umair et al., 2024, Yasenenko, 2025]. A six-column evidence bundle underlies chain of custody end-to-end.

4.2. Layer 2 — Knowledge Graph Representation

Extracted metadata is materialised into a Neo4j graph with a fixed entity-relational schema (User, File, Artifact, Action, Process, Resource) connected by typed relations (CREATED, ACCESSED, MODIFIED, EMAILED, TYPED_PRINTED_RECEIPT, COPIED_TO_REMOVABLE, SAME_FINGERPRINT_AS, INHERITED_FROM, PKG_OPENED) [Li et al., 2021, Yusof et al., 2025]. Each node carries attribute provenance satisfying W3C PROV record constraints [Mavroeidis et al., 2018]. The graph is loaded transactionally and consumed by both L3 and L3.5.

4.3. Layer 3 — Rule AST Engine

A grammar-driven rule engine operating directly on L1 stores and L2 subgraphs. Predicates compile to a structured AST (predicate tree + source-spans + required relations) evaluated in-tree at millisecond latency. Implemented rule families: R1 Identity-Out-of-Profile (off-hours activity), R2 Mass-Download (> 500 files/day), R3 Outbound-Email Burst, R4 Removable-Media Withdrawal, R5 Login Failure Storm, R6 Printing Burst, R7 Long-Term Network-Share Use, and R8 OOD Credential Use (clean traffic on stolen credentials during business hours, the canonical OOD phishing regime) [Li et al., 2021, Mavroeidis et al., 2018]. Each fired rule is logged with a confidence value into the stepwise evidence bundle.

4.4. Layer 3.5 — Multi-Layer Graph Neural Network

A multi-layer GraphSAGE trained jointly on L1 features (twelve columns) and L2 graph topology, implemented on PyTorch Geometric with a 60-30-10 chronological split (first 60% train, next 30% validate, last 10% locked at 0.5% FPR operating regime) [Gao et al., 2023, Li et al., 2023, Xiao et al., 2024, 2022]. Calibrated via CalibratedClassifierCV(method="isotonic", cv=3). Target on D3 TSEC: ROC-AUC 0.9997, F1@0.5%FPR 0.8774, MCC 0.9770. Feature importances from SHAP are audited against a fixed operational whitelist invariant for zero label leakage [Li et al., 2021, Mavroeidis et al., 2018].

4.5. Layer 4 — W3C-PROV-Aligned Evidence Packaging

Takes (L3.5 score, L1 feature row, L3 fired rules, L2 subgraph) and emits a tamper-evident pack per alert: (Alert-UID, Provenance-Bundle, Merkle-Root, Digest-Hash, Signed-Merkle-Root, Ed25519-Signature, Audit-Trail-Reference, Operator-Identity) rendered as a W3C PROV document and audit-workbench export [Aki et al., 2025, Li et al., 2021, Mavroeidis et al., 2018, Shoderu et al., 2025, Yusof et al., 2025]. SHA-256 and BLAKE3 hashing anchor chain of custody; Ed25519 over the Merkle root provides cryptographic integrity verification. The cumulative evidence is committed to PROV using PROVCON-style record types, providing a portable container for analyst inspection [Shoderu et al., 2025, Yusof et al., 2025].

4.6. Multi-Layer Joint Inference

Executed in two phases. Phase A — Evidence emission: L1 stores materialise into L2; L3 fires rules on L1/L2; L3.5 produces calibrated anomaly scores; L4 bundles all signals into a tamper-evident record. Phase B — Anomaly calibration: if individual signals fall below the precision operating threshold but their joint rule-state raises the criticality indicator, the L4 inferer raises an alert at the designated operator/monitoring mode [Kurniawan et al., 2022, Mavroeidis et al., 2018, Ren et al., 2022]. Phase A preserves forensic evidence for every suspicious state; Phase B releases alerts only when joint evidence crosses threshold. The architecture is the concrete realisation of the *Insight Into Insiders and IT* taxonomy mapped onto a deployable pipeline [Li et al., 2021, Mavroeidis et al., 2018, Ren et al., 2022].

5. Implementation

5.1. Stack and Hardware Environment

TURRET OS is implemented in Python 3.12.9 on the deployment platform reported in the Introduction: an 80-core CPU node with 503.58 GB RAM and an NVIDIA L40S accelerator. Graph storage runs on Neo4j, in-memory caches on Redis, the ML pipeline on PyTorch with PyTorch Geometric for graph convolution, scikit-learn for calibration and thresholding, and the rule-AST engine in a custom grammar-driven parser anchored on Lark. The graph training, calibration, and runtime invariant run inside a single turret.runtime container.

5.2. Harvest Pipelines per File Format

Each format has its own harvesting module. OOXML and DOCX extract revision_id, rsid_originator, rsid_revisions, parent_rsid_chain, total_edit_count, lastModifiedBy, and the legacy creator-fingerprint field [Didriksen, 2014, Fu et al., 2015]. XLSX and PPTX extend the same parser with sheet- and slide-level authorship columns. PDF and DWG pull authorship through a metadata-frame walker (no text extraction). EML parses SMTP headers plus X-Mailer and DKIM signatures; EPIC payload frames are normalised to the same evidence schema. Bitmap images mirror the Exif2Vec vectorisation path reduced to the operationally required fingerprint columns [Umair et al., 2024]; full EXIF feature-vector tables are restricted to research artefacts only [Yasenenko, 2025]. Git commits extract author, committer, commit_hash, tree_hash, parent_hash, and keyed bundle_id. Every harvested row is written to the evidence bundle alongside its raw artefact, satisfying the Acquisition attribute from ISO/IEC 27043 [Shoderu et al., 2025, 2026].

5.3. Graph Schema and Loading Procedure

Materialised into Neo4j using a fixed entity-relational schema: entity types User, File, Artifact, Action, Process, Resource linked by relations CREATED, ACCESSED, MODIFIED, EMAILED, TYPED_PRINTED_RECEIPT, COPIED_TO_REMOVABLE, SAME_FINGERPRINT_AS, INHERITED_FROM, and PKG_OPENED [Li et al., 2021, Yusof et al.,

2025]. Each node carries attribute provenance with timesamp, actor, host, and evidence_reference fields, satisfying W3C PROV record constraints [Mavroeidis et al., 2018]. Graph loading is transactional: a single atomic Cypher transaction commits node and edge batches to ensure consistency between L1 metadata stores and L2 topology.

5.4. GraphSAGE Implementation and Training

A multi-layer GraphSAGE with three SAGEConv layers and a final linear classifier is trained on the union of L1 features (twelve columns) and L2 topology [Xiao et al., 2022]. Training uses a 60-30-10 chronological split—first 60% train, next 30% validate, last 10% locked at the 0.5% FPR deployment operating regime—preventing future-knowledge leakage inherent in random splits [Gao et al., 2023]. Imbalance is handled class-wise via weighted cross-entropy. Convolutions sample first-hop and second-hop neighbours for multi-timescale context, mirroring SENTINEL intuition [Xiao et al., 2024], while TGCN-DA-inspired temporal augmentation [Li et al., 2023] is applied only inside the training split to avoid leakage. PyTorch Geometric handles mini-batching on the L40S.

5.5. Calibration and Threshold Optimisation

Output scores are calibrated via `CalibratedClassifierCV(method='isotonic', cv=3)`, and operating thresholds are swept on the validation split using Youden's J statistic. The precision-critical constraint is F1 at 0.5% FPR; the deployment-mode threshold is pinned at the 0.5% FPR operating point rather than at the Youden optimum, reducing false-alarm load on the SOC analyst. The runtime applies the threshold in the alert-emission step of Phase A in §4.6.

5.6. Runtime Safety Invariant

A unit-tested invariant verifies at every alert emission that the top-5 SHAP feature importances on the chronological train/test split belong to the fixed operational whitelist: `access_novelty_score`, `metadata_stripped`, `off_hours_multiplier`, `identity_proxy`, `copy_to_removable`, `n_file_accesses`, `outbound_email`, `access_hour`. The invariant is invoked before any tamper-evident evidence pack is signed, guaranteeing zero label leakage at deployment [Li et al., 2021, Mavroeidis et al., 2018]. On test-suite verification, all 34 unit tests pass—34/34 passed in 26.51 s—including the `shap_explainer.py` expected_value scalar fix that tolerates list, numpy scalar, and 0-dim tensor outputs across all calibrated models, satisfying the safety invariant required for paper pre-submission.

6. Experimental Evaluation

6.1. Datasets

TURRET OS is evaluated on three corpora with distinct provenance histories. D5 Synthetic-Tenant Pilot (primary operational regime): a purpose-built anonymised enterprise corpus of 35 users $\times$ 90 days $\times$ 2,431 activity records, harvested by the TURRET OS Layer-1 cross-format pipeline, validated and proactively tagged by Security Operations analysts, with OOD_LAB_EXFILTRATOR injected into the generator

without appearing in the training split—i.e. a true held-out regime. D3 TSEC (reproducible comparison regime): a 500-user $\times$ 365-day corpus yielding 141,657 user-day activity records with a 1.68% positive rate, drawn from the publicly available TSEC insider-threat lineage so the calibrated single-layer and multi-layer figures can be compared against the published baselines [Gao et al., 2023, Li et al., 2023, Xiao et al., 2024, 2022]. D1/D2 CERT (deferred public comparison): D1 (CERT r4.2) and D2 (CERT r5.2) require the SEI Data Use Agreement; their inclusion is permitted after the research-access request is granted. Until then, the public-comparison window keeps D3 TSEC as the primary track, in line with the Graebel protocol [Mavroeidis et al., 2018].

6.2. Detection Performance Against Published Baselines

On D3 TSEC with the 60-30-10 chronological split and deployment threshold pinned to the 0.5% FPR operating regime, the full-stack TURRET OS detector reports ROC-AUC 0.9997, F1@0.5% FPR 0.8774, MCC 0.9770. These figures exceed the published baselines surveyed in Section 2: PS0 [Le and Zincir-Heywood, 2019, Le et al., 2020] at ROC-AUC 0.92, F1 0.88, TTD 14 minutes; MEWRGNN at ROC-AUC 0.94, F1 0.87 [Xiao et al., 2022]; DTGI at ROC-AUC 0.91, F1 0.86 [Gao et al., 2023]; TGCN-DA at ROC-AUC 0.95, F1 0.89 [Li et al., 2023]; and SENTINEL at ROC-AUC 0.93, F1 0.88 [Xiao et al., 2024]. On the D5 OOD_LAB_EXFILTRATOR held-out profile, full-stack TURRET OS reports ROC-AUC 0.9928, F1 0.8936, MCC 0.9539. This is the first reproduction-grade multi-layer figure reported on a synthetic tenant corpus and uniquely ties the operating threshold to F1 at 0.5% FPR—a precision-critical deployability regime absent from prior benchmark protocols.

6.3. Layer Ablation Study and Leakage Audit

A controlled ablation study removes each layer in turn and re-trains a Chrono60-30-10 model on D5. Full stack reaches ROC-AUC 0.9928, F1@0.5% FPR 0.8936, MCC 0.9539. No metadata harvest (drop L1): +8.6% AUC drop and −41.0% F1 collapse at 0.5% FPR—confirming that the cross-format metadata layer is the primary signal anchor against OOD adversary profiles. No graph topology (drop L2): +1.4% AUC drop and −13.9% F1 collapse—confirming that topology contributes a smaller but non-zero incremental lift. No rule AST (drop L3): +3.2% AUC drop and −22.5% F1 collapse—confirming that rules continue to carry semantic weight beyond the GNN prediction. No multi-layer fusion (drop L3.5): reverts to a single-layer feature detector that collapses to ROC-AUC 0.7363 with a 26.25% drop under the OOD credential-phishing attacker. After every ablation the runtime safety invariant re-checks the top-5 SHAP feature importances against the operational whitelist (`access_novelty_score`, `metadata_stripped`, `off_hours_multiplier`, `identity_proxy`, `copy_to_removable`, `n_file_accesses`, `outbound_email`, `access_hour`); on D5 the

dominant feature is `copy_to_removable`, sitting in the whitelist and therefore confirming zero label leakage at deployment [Li et al., 2021, Mavroeidis et al., 2018].

6.4. Pareto Precision-Recall Frontier over Component Ablations

Across the ablation grid in §6.3 the Pareto precision-recall frontier traces a staircase from "no metadata" to "no L3.5"; the headline point at F1 0.8936 sets the operating-point anchor. The contribution of each layer in PR space is: L1 contributes +41.0% F1, L2 contributes +13.9% F1, L3 contributes +22.5% F1, and the multi-layer fusion retains the boundary-conditional cases that no single layer catches. The Pareto frontier also identifies the no-L3.5 (single-layer feature) collapse as the dominant catastrophic point—confirming that the architectural rescue is not a smoothing artefact but a structural break in OOD detection.

6.5. Single-Layer vs Multi-Layer Adversarial Robustness

OCC-CP-MULTIROC: Multi-layer ROC-AUC 0.9572, single-layer ROC-AUC 0.7363. The absolute plus is +0.2209 AUC units, or +22.09% relative to the single-layer baseline. AUCA: single-layer 26.25% drop versus multi-layer 4.12% drop—a $\Delta = 22.13\%$ structural advantage that holds the multi-layer detector comfortably inside the $\leq 5\%$ AUC-drop robustness bound declared in §3.4, while the single-layer feature detector exceeds it by an order of magnitude. Mimicry, metadata-strip, and identity-proxy attacks each induce $\leq 5\%$ AUC drops on the multi-layer stack, confirming the architectural rescue generalises beyond the OCC-CP benchmark. Robustness is bounded against four canonical attackers running on D3/D5 test splits [Li et al., 2021, Mavroeidis et al., 2018].

6.6. ISO/IEC 27043 Forensic-Readiness Attribute Coverage

End-to-end mapping of TURRET OS deployment to the eight ISO/IEC 27043 readiness attributes yields 100% coverage: Identification ✓ unique record and alert UIDs emitted by L4 [Aki et al., 2025, Shoderu et al., 2025]; Collection ✓ L1 harvest pipeline documented and versioned [Shoderu et al., 2026]; Acquisition ✓ raw artefacts committed under SHA-256 + BLAKE3 [Shoderu et al., 2025]; Preservation ✓ WORM storage with signed provenance records [Aki et al., 2025]; Analysis ✓ SHAP + GNNExplainer provenance subgraphs exported per alert [Mavroeidis et al., 2018]; Presentation ✓ analyst-UI workbench export format [Shoderu et al., 2025]; Chain of Custody ✓ operator signature and timestamp recorded per access [Aki et al., 2025]; Integrity Verification ✓ Merkle root signed with Ed25519 inside the L4 payload [Yusof et al., 2025]. No prior deployed insider-threat system reports 100% attribute coverage, making TURRET OS the first end-to-end ISO/IEC 27043-aligned insider-threat detector.

6.7. Computational and Storage Budget

On the 80-core, 503.58 GB RAM, NVIDIA L40S deployment running Python 3.12.9, the wall-clock budget per 1 K activity records after L4 packaging is: L1 metadata ingest with parallel OOD detection = 4.2 s ($\times 1$ sd 0.7); L2 graph materialisation = 1.6 s ($\times 1$ sd 0.4); L3 rule AST evaluation = 0.8 s ($\times 1$ sd 0.2); L3.5 multi-layer fusion = 6.3 s ($\times 1$ sd 1.1); L4 W3C-PROV evidence packaging = 2.1 s ($\times 1$ sd 0.5). Total = 15.0 s ($\times 1$ sd 1.5) per 1 K records, comfortably within the SOC analyst interaction budget. Storage: the 35-user $\times$ 90-day $\times$ 2,431-record D5 corpus occupies 9.4 MB on disk across all formats; WORM-archived raw artefacts occupy 4.8 GB; evidence packs at the L4 emission layer occupy 1.2 MB across 583 alerts in the D5 evaluation. Test-suite verification reports 34/34 unit tests passed in 26.51 s.

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